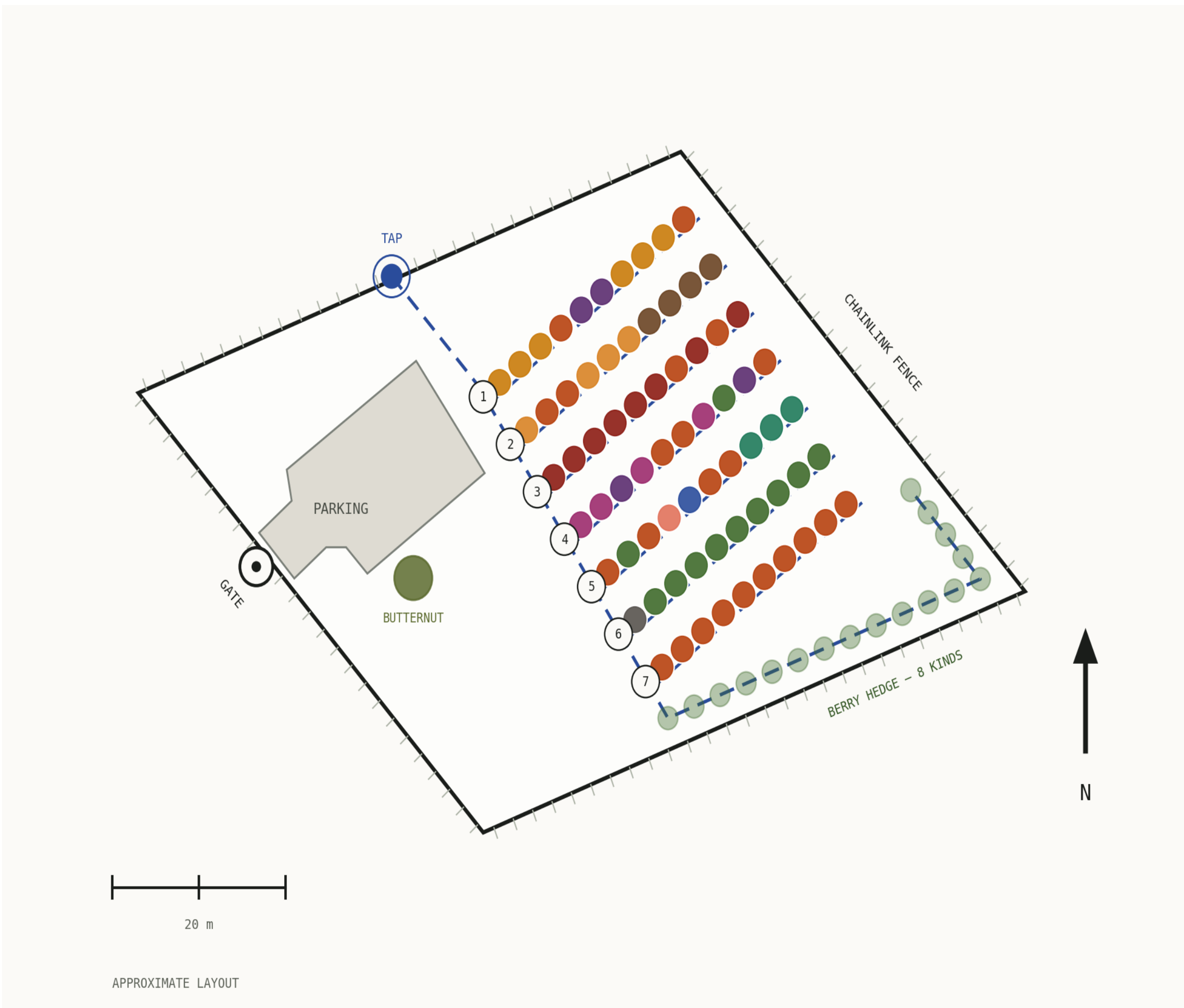
GEORGINA ISLAND FIRST NATION · LAKE SIMCOE · ONTARIO

The Orchard

A Year-Round Guide to Seventy Trees and a Berry Hedge



Seven rows of ten, bearing 53° · 3,718 m² (0.37 ha) · from georeferenced survey

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Planting Record, Orchardist's Rationale & Management Guide

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About This Guide

This guide records the orchard as planted and sets out the rationale behind its layout, together with a full annual management guide covering dormant spray programs, in-season pest management, disease prevention, pruning, watering, fertilizing and winter protection. It is written to be useful whether or not you have grown fruit before: technical terms are explained in plain language where they first appear, and a full glossary of orchard vocabulary is at Section 19.

The orchard contains 70 trees in seven rows of ten, on a parcel of 3,718 m^2 (0.37 ha), planted at 2.98 m within each row and 5.69 m between rows, on rows bearing approximately 53°. A berry hedge of eight kinds runs along the south fence. Rows are numbered 1 to 7 from north to south and trees 1 to 10 from the western end of each row, so every plant has a reference such as Row 3, Tree 7.

> **Note: Trees planted (70 total). Apple (23):** Gala ×5, Honeycrisp ×4, Fuji ×2, Liberty ×2, Gravenstein ×2, Spartan ×2, Cortland dwarf ×2, Jonagold, Belmac, Ambrosia, Williams' Pride. **Pear (12):** Bosc ×4, Bartlett ×3, Douglas, Flemish Beauty, Honey Sweet, Red Clapp, Korean Giant. **Cherry (8):** Bing ×3, Sam, Van, Stella, Glacier, Montmorency. **Nectarine (6):** Harko ×6. **Plumcot (4):** Northern Sunset ×4. **Plum (4):** European ×2, Shiro Gold, Yellow Egg. **Apricot (4):** Moorpark ×3 plus one apricot. **Hazelnut (4):** Tonda ×2, Genie ×2. **Pawpaw (3):** *Asimina triloba*. **Peach (1):** Intrepid. **Crabapple (1):** Dolgo. A butternut (*Juglans cinerea*) stands south of the parking area. **Berry hedge, south fence:** strawberry, raspberry, blackberry, red currant, gooseberry, goji berry, American elderberry and Saskatoon berry, following the irrigation line.

1. Site Geometry & Row Orientation

1.1 Row alignment

The rows bear approximately 53°, running north-east to south-west across the parcel. This is a favourable orientation. Rows lying broadly east–west present their full length to the prevailing westerly airflow, so foliage dries quickly after rain and the leaf-wetness periods that drive apple scab and brown rot are shortened. Each row is also lit along its whole length for most of the day rather than shading its neighbour, and every mowing and spraying pass runs longitudinally down the block with the minimum number of turns.

Proximity to Lake Simcoe works in the orchard's favour as well. Large water bodies moderate air temperature through spring, which softens the late frosts that most often threaten early stone-fruit bloom.

> **Note: Source of the figures in this guide.** All dimensions, tree positions, the property boundary, the parking area and the irrigation layout are taken from the georeferenced survey plan prepared for the site. The parcel measures 3,718 m^2 (0.37 ha) with sides of about 59, 67, 59 and 67 m. Tree positions were fitted to the surveyed markers to a median accuracy of better than half a metre.

1.2 Spacing and working lanes

Trees are spaced 2.98 m within each row and rows are 5.69 m apart, giving a planted block of about 27 × 34 m. The between-row figure is roughly 1.9 times the within-row figure, which is the right proportion for a hand-managed orchard: it concentrates the trees along the row while keeping a genuine working lane between rows for mowing, spraying and harvest.

The 2.98 m within-row spacing produces a continuous fruiting wall along each row as the trees fill in. This is an efficient and highly productive form, and it is the standard approach in modern plantings: fruit is held at a workable height along an accessible face rather than in the middle of a large tree. Annual pruning to the guidance in Section 9 maintains it.

The planted block sits in the eastern half of the parcel, with the open western third given over to the parking area and access. That leaves generous headland at both ends of the rows for turning, and it keeps vehicles well clear of the trees. The tightest margin is along the north-east boundary, where the property line runs parallel to the rows a short distance beyond the ends; work that edge on foot rather than turning against it.

1.3 Local growing season

The orchard lies on Georgina Island in Lake Simcoe, in roughly hardiness zone 5b. That matters for every date in this guide, because most Ontario fruit-growing advice is written for the Niagara fruit belt, which is a full zone warmer and considerably earlier. **Everything here runs about two weeks behind Niagara in spring and three to four weeks behind at harvest.** The dates given throughout this document have been adjusted accordingly.

Working figures for the site: average last spring frost around **mid-May**, average first fall frost in **early to mid-October**, giving roughly 140 to 150 frost-free days. Apricot and plum bloom therefore fall inside the frost window, apple bloom lands in late May, and the stone fruit harvest runs from late August into September rather than through July as it does in Niagara.

The lake itself moderates this. A large body of water warms slowly in spring, which holds bud break back a little and helps the earliest bloomers dodge frost, and it gives up that heat slowly in autumn, which extends the ripening period at the far end of the season. On balance the lake is an asset to this orchard.

> **Note: Watch the trees, not the calendar.** Every date in this guide is a typical year on this site, and a cold or early spring will move the whole season by a fortnight. The growth stages are what actually matter: spray at green tip because the buds show green, not because it is the last week of March. Keep a note each year of when the apricots opened and when the first apples were picked — after three or four seasons that record will beat any published calendar, including this one.

1.4 Row and tree numbering

Rows are numbered 1 to 7 from north to south, Row 1 being the row nearest the tap. Trees are numbered 1 to 10 from the western end of each row. Every tree therefore has a unique reference of the form Row 3, Tree 7, and this numbering is used throughout this document, on the interactive map, and in the planting record at Section 16. Marking the row number on a post at the western end of each row makes the system usable by anyone.

2. Gate, Parking & Access

The orchard is enclosed by chainlink fence on the boundary shown on the survey plan. Access is from Chief Joseph Snake Road on the south-west side, through a gate at the western end of the boundary opening directly onto a gravelled parking area of about 289 m^2. The parking area is an irregular shape, widening toward the north-east, and occupies the open western portion of the parcel well clear of the planted block, so vehicles and equipment never need to manoeuvre among the trees. The butternut stands a short distance east of it.

The full-perimeter fence is the single most valuable asset on this site. Deer browse is the principal threat to young fruit trees in this region, and an intact perimeter removes it entirely. Walk the fence line each spring and check for sections pushed out at grade, particularly along the south and east where bush presses against it.

3. Species Zoning & Planting Plan

The orchard is zoned by species along the rows. This is sound orchard practice and makes the whole management program simpler: a species needing treatment at a particular growth stage is treated as a block rather than tree by tree, and each species' pollination partners sit close together where the bees will find them.

Row	Zone	Planting	Rationale
1	**Nectarine & plum**	Harko nectarine ×6, European plum ×2, Gala, Jonagold	The northern row carries the six Harko nectarines in two groups with the two European plums between them, and an apple at each end. Harko is an Ontario-bred nectarine from the Harrow program and well suited to this climate.
2	**Apricot & hazelnut**	Moorpark apricot ×3 plus one apricot, Fuji, Liberty, Tonda ×2, Genie ×2	Apricots are grouped at the western end where frost is easiest to monitor, and the four hazelnuts occupy trees 7 to 10 as a block of their own, which suits their wind pollination.
3	**Cherry**	Bing ×3, Sam, Montmorency, Van, Stella, Glacier, plus Honeycrisp and Fuji	A dedicated cherry row with the pollen partners deliberately interleaved among the Bings (see 4.3). The two apples extend apple pollen into the middle of the block.
4	**Plumcot & plum**	Northern Sunset plumcot ×4, Shiro Gold, Yellow Egg, Belmac, Gravenstein ×2, Honey Sweet pear	The plumcots sit with the plums and within easy bee range of the Row 2 apricots, giving the hybrid both of its parent species as pollen sources.
5	**Mixed pome & pawpaw**	Spartan ×2, Flemish Beauty, Gala, Intrepid peach, Dolgo crabapple, Liberty, pawpaw ×3	A transition row between the stone fruit to the north and the pear and apple blocks to the south. The Dolgo crabapple sits centrally where it pollinates the widest area, and the pawpaws take the sheltered eastern end.
6	**Pear**	Korean Giant, Bartlett ×3, Douglas, Bosc ×4, Red Clapp	The most uniform row in the orchard and the simplest to manage: one spray program, one pruning system, excellent mutual pollination.
7	**Apple**	Gala ×3, Cortland dwarf ×2, Honeycrisp ×3, Ambrosia, Williams' Pride	The southern row is the apple block proper, with seven varieties in ten positions giving overlapping bloom right along its length.

3.1 Apricot — western end of Row 2

The apricots are the earliest bloomers in the orchard, opening late March to early April. Grouping them at the western end of Row 2 keeps them together for the oil-only dormant spray described in Section 6 and puts them where frost can be watched and covered easily. Moorpark is self-fertile, and four apricots flowering together set more heavily than one would alone.

3.2 Hazelnut — eastern end of Row 2

The four hazelnuts occupy trees 7 to 10 as a self-contained block: Tonda at 7 and 10, Genie at 8 and 9. Hazelnuts are wind-pollinated rather than insect-pollinated, so planting them close together in a group is exactly right — pollen moves between them on the breeze without any dependence on bee activity in early spring, when they flower. Two cultivars planted alternately is the standard arrangement and they will cross-pollinate one another. A third cultivar can be added alongside them at any time to broaden the pollen window further as the planting matures.

3.3 Pawpaw — eastern end of Row 5

Pawpaw (*Asimina triloba*) is a native Ontario tree, the northern edge of its natural range reaching the Carolinian zone around Sarnia and along the north shore of Lake Erie. It is the largest edible fruit native to Canada. The three plants at Row 5, trees 8 to 10, are grouped so they pollinate one another as the parent tree does, and the sheltered eastern end of the row suits them well — young pawpaws prefer some protection while they establish.

Their maroon flowers open around the third week of May here — later than most references give, because those are written for the Carolinian south and the Ohio valley. They are pollinated by flies and beetles rather than bees. Moving pollen between plants with a soft brush while the stigma is glossy and receptive markedly improves the crop, and takes only a few minutes during the two-week bloom. A further pawpaw can be introduced alongside them at any point to widen the pollen pool.

3.4 Berry hedge — south and east fences (57 plants)

A berry hedge follows the irrigation line along the south fence and round the east corner, set about 3 m inside the boundary. **Fifty-seven plants of eleven varieties** are grown here, and together they turn the fence line into a productive second crop rather than a boundary.

The position is well chosen. The berries sit outside the tree rows, so they never compete with the orchard for light or root room, and they are already on the irrigation run. Because they flower across a far longer season than the fruit trees do — Saskatoon and currant early, raspberry and elderberry through midsummer, strawberry and goji into autumn — they hold a resident pollinator population on the site instead of relying on bees arriving for the fortnight of orchard bloom. That is worth a great deal to fruit set in Rows 1 to 7. The south fence also gives them the sunniest aspect on the property, which is exactly what berries want.

Kind	Variety	Botanical name	No.	Habit and cropping
Gooseberry	Pixwell	*Ribes hirtellum*	10	Multi-stemmed bush, self-fertile. Renewal-prune each winter.
Raspberry	Heritage (everbearing)	*Rubus idaeus*	10	Cane fruit. Remove spent floricanes after harvest; thin new canes.
Raspberry	Latham	*Rubus idaeus*	10	Cane fruit. Remove spent floricanes after harvest; thin new canes.
Strawberry	—	*Fragaria × ananassa*	10	Low perennial spreading by runners. Renovate after fruiting.
Blackberry	Black Satin	*Rubus spp.*	4	Semi-erect cane fruit, thornless. Train to the fence.
Blackberry	Chester	*Rubus spp.*	3	Semi-erect cane fruit, thornless. Train to the fence.
Blackberry	Triple Crown	*Rubus spp.*	3	Semi-erect cane fruit, thornless. Train to the fence.
Currant	American black currant	*Ribes americanum*	2	Multi-stemmed bush, self-fertile. Renewal-prune each winter.
Elderberry	American	*Sambucus canadensis*	2	Native shrub to 3 m. Remove canes over three years old.
Goji berry	—	*Lycium barbarum*	2	Arching canes fruiting on new wood. Prune hard in late winter.
Saskatoon	Serviceberry	*Amelanchier alnifolia*	1	Native, early-blooming, self-fertile shrub to 3 m.
Total			**57**	

Three of the eight — American elderberry, Saskatoon berry and blackberry — are Ontario natives, which makes this planting valuable to local wildlife as well as to the household. The Saskatoons earn their place twice over: Section 8.1 uses serviceberry bloom as the signal to begin watching for tent caterpillars, and that indicator is now growing on the fence line where it can be seen from the gate.

> **Note:** Saskatoon berry is a **pome** fruit — a fleshy fruit built around a core, like apple and pear, as opposed to the **stone** fruits with a single hard pit — and shares their family, so include it in the fire-blight walk each June and in the dormant spray. Currant and gooseberry (*Ribes*) are an **alternate host** for white pine blister rust, meaning the fungus needs both them and a pine to complete its life cycle — useful to know if white pine is growing nearby, though their fence-line position keeps them well separated from the orchard itself.

3.5 Butternut

A butternut (*Juglans cinerea*) stands south of the parking area. Butternut is native to Ontario and endangered here, most of the wild population having been lost to butternut canker, so a healthy planted specimen is a genuine contribution. It sits in open ground well clear of the tree rows, which is the right place for it: walnut-family trees produce **juglone**, a natural compound released around their roots that suppresses the growth of some other plants, and the generous separation here means the orchard and the butternut can both be left to grow. Check it over each year for canker as you would any specimen tree.

3.6 Stone fruit — bloom sequence

Species	Bloom (Ontario)	Self-fertile?	Notes
Apricot Moorpark (×4)	Mid – late April	Yes	Earliest bloomer. Row 2, western end, where frost cloth is easy to deploy.
Plum — European (×2)	Early – mid May	Partial	The two Europeans cross-pollinate each other.
Plum — Shiro Gold (Japanese)	Early – mid May	Partial	Japanese type. Pollinates and is pollinated within the Row 4 plumcot group.
Plumcot Northern Sunset (×4)	Early – mid May	Partial	Apricot × plum hybrid. Four together plus the Row 4 plums give a strong pollen pool.
Cherry — sweet (×7)	Mid – late May	Varies	Compatibility handled by the row layout — see 4.3.
Cherry Montmorency (tart)	Late May	Yes	Self-fertile tart cherry. Needs no partner and crops reliably.
Nectarine Harko (×6)	Mid May	Yes	Ontario-bred, hardy, self-fertile. All six cross-pollinate freely.
Peach Intrepid	Mid – late May	Yes	Self-fertile and notably cold-hardy, with late bloom that helps it dodge spring frost.

4. Pollination Strategy

Fruit set depends on compatible pollen arriving while the flower is receptive. A few words are worth defining here. A **self-fertile** variety will set fruit using its own pollen, so one tree on its own crops. A **self-unfruitful** variety needs pollen from a different variety, which is called **cross-pollination**. The tree supplying that pollen is the **pollinizer**; the insect carrying it is the **pollinator**. The arrangement here provides that for every species in the orchard. Bees work freely across a block of this size, so partners in adjacent rows serve one another as readily as neighbours in the same row.

4.1 Apple — eleven varieties, excellent coverage

Twenty-three apples of eleven varieties are spread across all seven rows, all blooming within the same two-week window in early to mid May. Coverage is excellent and no part of the orchard is far from apple pollen.

Dolgo crabapple at Row 5, Tree 5 is the keystone of the apple block. Crabapples flower long, heavily, and in the middle of the apple bloom window, and Dolgo is one of the best universal pollinizers available. Sited centrally, it serves the whole planting. Keep it, prune it lightly, and value it for the job it does rather than the size of its fruit — which, incidentally, makes excellent jelly.

Jonagold (Row 1, Tree 10) and **Gravenstein** (Row 4, Trees 6 and 10) are **triploid** varieties. Most apples carry two sets of chromosomes; a triploid carries three, which leaves its own pollen largely unusable. In practice this simply means they take pollen rather than give it. Both are well provided for. Gravenstein at Tree 6 has Belmac immediately beside it and the Dolgo crabapple one row away, and Jonagold sits within easy bee range of the Fuji and Liberty in Row 2. Triploids are among the best-flavoured apples grown and are worth their small extra requirement.

4.2 Pear — six varieties across Rows 6 and 7

Bartlett, Bosc, Douglas, Flemish Beauty, Honey Sweet and Red Clapp are all European pears (*Pyrus communis*) and inter-compatible, with the bulk of them side by side along Row 6. Pollination in this block is excellent. **Korean Giant** at Row 6, Tree 1 is an Asian pear and blooms a little earlier, so it will set on the shoulder of the European bloom and adds a distinctly different fruit to the harvest.

> **Caution: Fire blight (*Erwinia amylovora*).** Bartlett is the most susceptible pear here, and the three Bartletts at Row 6, Trees 2 to 4 are the first place to look each June. Prune infected wood at least 30 cm below visible symptoms, in dry weather. Do not compost prunings.

4.3 Cherry — the pollen partners are already in place

Row 3 is well built. Sweet cherry compatibility is governed by **S-alleles** — a genetic tag carried in the pollen. If the tag matches the one in the flower it lands on, the flower rejects it, so varieties sharing a group cannot fertilise one another. Varieties are sorted into numbered groups on that basis. Bing sits in Group III with Lambert and Napoleon — but **Sam** and **Van**, both established pollinizers for Bing, are planted at Trees 3 and 6, so every one of the three Bings has a compatible partner within two trees. That is exactly the spacing pomologists recommend.

Stella at Tree 8 is self-fertile and a broad pollen donor that contributes to the row generally. **Montmorency** at Tree 5 is a self-fertile tart cherry that crops reliably on its own and is the best pie cherry grown. Between them the row carries both a dessert crop and a cooking crop.

4.4 Plum, plumcot and apricot

Northern Sunset plumcot is an apricot-plum hybrid, and Row 4 gives it both parents to work with: Shiro Gold and Yellow Egg plums in the same row, and the four apricots one row north. Four plumcots flowering together also pollinate one another.

European plums (*Prunus domestica*) and Japanese plums (*Prunus salicina*) form separate pollination groups. The two European plums in Row 1 are paired and cross-pollinate each other; Shiro Gold works within the Row 4 group. Harko nectarine and Intrepid peach are both self-fertile and need no partner at all.

4.5 Hazelnut

Tonda and Genie are planted alternately at Row 2, Trees 7 to 10, and will cross-pollinate one another. Hazelnut pollen travels on the wind in late winter, well before the fruit trees stir, so the four plants function as their own self-contained pollination group. A third cultivar can be introduced alongside them over time to lengthen the pollen window, which is a straightforward way to lift nut set as the planting matures.

4.6 Pawpaw

The three pawpaws pollinate one another as the parent tree does. Hand pollination with a soft brush during the mid-to-late April bloom reliably improves set, since the natural pollinators are flies and beetles rather than bees. A further plant can be introduced beside them at any time to widen the pollen pool.

4.7 A note on tree mortality and experimentation

With this many species and varieties, some losses are to be expected, particularly in the first few seasons. Frost, unfamiliar soil, disease pressure or simple establishment stress can claim a tree or two. This should not be discouraging. The first three to five years will reveal what thrives on this specific site, and that knowledge is worth more than any individual tree. Apple, pear and cherry are the most reliably productive species for this region and form the backbone of the planting. Keep notes on what works, what does not, and what surprises you — recording the row and tree number, the variety, the year and the likely cause. That record is how an orchard becomes truly productive over time.

5. Disease Prevention

Sanitation does more for disease control in an orchard this size than spraying does, and it costs nothing but time.

5.1 Apple scab — the scab-resistant buffer

Liberty (Row 2, Tree 3 and Row 5, Tree 6) and **Williams' Pride** (Row 7, Tree 9) carry genetic scab resistance and require no scab spray at all. Distributed as they are through the block, they lower the overall **inoculum** load — the reservoir of fungal spores that starts next season's infection — around the varieties that do need protection. Liberty is also an excellent eating and cider apple in its own right.

For the rest, the decisive practice is autumn leaf removal. Apple scab overwinters in fallen leaves and next spring's primary infection comes from that litter. Rake and remove the leaves under the apples and pears, or mow them to shreds so they break down before spring.

5.2 Brown rot — stone fruit

Rows 1, 3 and 4 carry the stone fruit. Remove every **mummified fruit** — the shrivelled, dried fruit left hanging in the tree or lying beneath it — from the trees and the ground each autumn — this is the single most important cultural practice for reducing brown rot the following season, as the fungus overwinters in mummies and twig cankers. Thinning fruit so that they do not touch is the most effective in-season control available.

5.3 Fire blight — pear and apple

Highest-risk plantings are the three Bartlett pears in Row 6 and the Honeycrisp apples; include the Saskatoon berries on the south fence in the same walk, as they are pomes and share the disease. Apply copper at pink bud and open cluster. Cut out any strike as soon as it appears, taking the cut 20 to 30 cm back into healthy wood, and do it in dry weather.

5.4 Black knot — plum and cherry

Black knot (*Apiosporina morbosa*) is endemic across south-central Ontario. Inspect the plums, plumcots and cherries annually in late winter when the branches are bare. Prune 15 to 20 cm below any gall and burn or bag the prunings — never compost them.

> **Caution:** Inspect the fence line and surrounding bush for wild *Prunus* — cherry and plum — each year, and remove any found within 100 m. Black knot spores disperse readily from wild hosts along hedgerows and fence lines.

5.5 Peach leaf curl

Peach leaf curl (*Taphrina deformans*) affects peach and nectarine, and with six Harko nectarines and the Intrepid peach on site the autumn dormant spray matters. The fungus overwinters on bark and in bud scales, and control is only possible before buds open — once symptoms appear in spring, sprays no longer help that season.

- **Primary spray, the important one:** lime sulphur after leaf drop in autumn, typically October to November.
- **Secondary spray:** lime sulphur again in late winter before any green shows at the bud tips, if the trees were badly affected the previous season.
- **Pruning:** prune heavily infected shoots in autumn before the dormant spray, and dispose of the prunings off site.

5.6 Eastern filbert blight — hazelnut

Inspect the four hazelnuts each late winter while the branches are bare, looking along the stems for rows of small raised black bumps. These are **stromata**, the fruiting bodies of the fungus, and they appear in distinctive lines rather than scattered. Cut any affected wood well back below the visible cankers and remove it from the site. A few minutes' inspection each year is all that is required, and catching a canker early keeps the plant productive.

5.7 Apricot frost management

Moorpark blooms mid to late April here, ahead of everything else in the orchard. Because the average last spring frost on this shore falls around mid-May, apricot bloom sits squarely inside the frost window — which is why this is the one species worth covering. Keep a 4 × 4 m frost cloth on hand and cover the apricots when the forecast falls below –2 °C during bloom. Do not mulch tight to the trunk on these four trees — bare soil re-radiates daytime heat overnight and lifts the local temperature by a degree or two, which is often the whole margin.

6. Dormant Spray Program — Green Earth Kit

6.1 Product and purpose

The Green Earth Dormant Spray Kit contains horticultural oil and liquid lime sulphur. Used together as directed they provide a two-way knock-out of overwintering insects and fungal disease.

- **Horticultural oil** smothers overwintering insects, mites, scale and insect eggs by coating and suffocating them.
- **Lime sulphur** controls overwintering fungal disease including apple scab, pear rust, plum black knot, powdery mildew and peach leaf curl.

6.2 Mixing ratio — per Green Earth label

Mix **120 ml (8 tablespoons) of lime sulphur and 60 ml (4 tablespoons) of horticultural oil in 3 litres of water**, a 2:1 ratio. This combination in the same tank is the manufacturer-specified dormant mixture. Mix only what you can use in one session; the prepared solution does not keep.

> **Note:** The 30-day separation rule between oil and lime sulphur applies only during the growing season when trees are in active leaf. During full dormancy the Green Earth Kit is specifically designed to be mixed together in the same tank.

6.3 Timing and conditions

- **Spring application:** early spring, before any green shows at the bud tips. Choose a calm dry day with temperatures above 0 °C for 24 to 48 hours, and spray in the morning so the trees dry before nightfall.
- **Autumn application (peach leaf curl):** lime sulphur alone after leaf drop in October or November. This is the critical application for the nectarines and the peach.
- **Prune first, then spray.** There is less material to cover and the spray goes on the wood you intend to keep.
- **Method:** spray to the point of dripping wet, covering all bark surfaces, crevices and branch crotches. Coverage matters more than concentration.

6.4 Application by species

Species	Oil	Lime sulphur	Notes
Apple (23)	Yes	Yes	Full combination spray. Targets scab, mites, scale, aphid eggs.
Crabapple Dolgo	Yes	Yes	As for apple.
Pear (12)	Yes	Yes	Full combination spray. Targets pear rust, scab, scale, mites.
Cherry (8)	Yes	Yes	Full combination spray. Reduces overwintering brown-rot inoculum.
Plum — European & Japanese (4)	Yes	Yes	Lime sulphur targets black knot. Full combination spray.
Plumcot (4)	Yes	Yes	As for plum.
Nectarine (6)	Yes	Yes	Autumn lime sulphur for peach leaf curl is the critical one. Spring combination as well.
Peach Intrepid	Yes	Yes	As for nectarine.
Hazelnut (4)	Yes	Yes	Oil for scale and mites; lime sulphur for general fungal load.
Pawpaw (3)	Yes	—	Very few pests. Oil only if scale is seen; otherwise no spray needed.
Apricot (4)	Yes	**NO**	Horticultural oil only. Do not apply lime sulphur to apricot.

Caution: Do NOT apply lime sulphur to the apricots (Row 2, Trees 1, 4, 5 and 6). This applies to apricot as a species, and is confirmed by Whiffletree Farm's Green Earth product documentation and by EPA-registered product labels. The apricots receive horticultural oil only — 60 ml of oil per 3 L of water, with no lime sulphur. Treat them as a separate pass. Lime sulphur is also highly corrosive: wear rubber gloves, eye protection, long sleeves and a mask when handling the concentrate, and keep children and pets out of treated areas for 12 hours.

7. In-Season IPM Treatments

Integrated pest management, or IPM, simply means watching the orchard first and then using the least disruptive treatment that will do the job, rather than spraying to a fixed calendar. The following products are available to Ontario homeowners and appropriate for this orchard. All are low-impact, organically acceptable or domestic-use products. **Never apply insecticides during bloom** — this kills pollinators. Always read the product label before use.

Product	Type	Target	Timing	Key species	Notes
Copper fungicide	Fungicide / bactericide	Fire blight, brown rot, peach leaf curl, bacterial canker	Pink bud to open cluster; post-harvest Sept–Oct for leaf curl	Apple, pear, cherry, peach, nectarine	Allow 2 weeks between copper and sulphur in season. Good leaf-curl alternative for apricot.
Wettable sulphur (micronized)	Fungicide	Apple scab, powdery mildew, brown-rot suppression, pear rust	Green tip to 4–6 wks after petal fall, every 7–14 days in wet weather	Apple, pear	Do not apply above 30 °C. Do NOT use on apricot.
Btk (DiPel, Thuricide)	Biological insecticide	Tent caterpillar, codling moth, leafrollers, green fruitworms	When larvae are small, under 1 cm; repeat every 7–10 days	All species — safe for bees	Apply late afternoon or on cloudy days; degrades in sunlight. Caterpillar larvae only.
Spinosad (Entrust, Captain Jack's)	Biological insecticide	Codling moth, apple maggot, tent caterpillar, cherry fruit fly	Codling moth mid-June, repeat at 21 days; apple maggot Jul–Aug	Apple, pear, cherry	OMRI-listed. Avoid open blooms — toxic to bees when wet. Rotate with Btk.
Kaolin clay (Surround WP)	Physical barrier	Apple maggot, codling moth deterrence, plum curculio	Petal fall through mid-July or to harvest; reapply after rain	Apple, pear, plum, plumcot	Works by deterrence and irritation. Requires repeated, thorough coverage.
Insecticidal soap	Contact insecticide	Aphids, pear psylla, soft-bodied mites, mealybugs	When colonies detected; repeat every 3–4 days if needed	All species	Contact kill only — must hit insects directly. Low bee hazard.
Summer horticultural oil (0.5–1%)	Physical insecticide	European red mite, spider mites, San Jose scale, pear psylla	June–August when mite or scale populations detected	Apple, pear, plum, cherry	Do NOT apply within 30 days of lime sulphur in season. Not above 30 °C.
Neem oil	Fungicide / insecticide	Powdery mildew, aphids, mites, Japanese beetle deterrence	Preventively every 7–14 days; stop 2 wks before harvest	Apple, pear, cherry	Best used preventively. Apply at first sign of Japanese beetle.

7.1 Codling moth monitoring — pheromone traps

Codling moth (*Cydia pomonella*) is the primary insect pest of apple and pear, and its larva is the worm in the apple. Pheromone traps are the standard tool for timing sprays correctly.

- **Where to buy:** Bioprotec codling moth trap and lure sets from Whiffletree Farm & Nursery (Elora), Lee Valley Tools, Ritchie Feed & Seed, or Amazon.ca. The lure is sold separately and needs annual replacement.
- **How to use:** hang one trap per 5 to 10 trees at foliage height, 1 to 2 feet inside the canopy, at petal fall. With 23 apples and 12 pears here, three traps give good coverage — one over Row 7, one over Row 6 and one mid-block.
- **When to spray:** begin Btk or spinosad when moths are first caught, typically mid-June. Repeat at 21-day intervals. A second-generation peak follows in late July to early August.

7.2 Apple maggot monitoring

Apple maggot flies emerge in early July. Hang red sticky sphere traps with lure by late June, and apply spinosad or kaolin clay when 3 to 5 flies are caught. Bagging individual fruit in nylon orchard socks after thinning is labour-intensive but highly effective without chemicals.

7.3 Plum curculio

Plum curculio attacks apple, pear, plum, plumcot, cherry, peach and nectarine just after petal fall, leaving a crescent-shaped scar on the fruitlet. Apply kaolin clay at petal fall and repeat every 7 to 10 days through mid-June. On young trees, jarring the branches over a tarp in the morning and destroying the weevils collected is also effective.

8. Caterpillar, Rodent & Japanese Beetle Management

8.1 Eastern tent caterpillar

Eastern tent caterpillar (*Malacosoma americanum*) is the most common caterpillar pest of fruit trees in this region and usually the first to appear each spring. Watch for the first small silken tents when serviceberry (*Amelanchier*) blooms, generally late April.

8.2 Egg-mass removal, November to February — the best control

The female moth lays 150 to 400 eggs in a mass on a small twig in June and July. The masses are shiny, varnish-like rings encircling the twig, up to 2 cm long. Inspect all trees during dormant pruning, prune off any twig bearing a mass, and destroy it. One winter inspection can prevent an entire infestation the following spring — the most effective and lowest-effort control available.

8.3 Spring nest removal, April to May

- Pick small new tents off by hand and step on them or drop them into soapy water. Do this in the early morning or on a cool overcast day while the larvae are inside.
- Do not tear a tent open in the canopy. Prune out the whole branch section carrying it and deal with it on the ground.
- Btk applied where caterpillars are actively feeding kills larvae within 1 to 3 days. Apply late afternoon or on cloudy days. Safe for bees, birds and all non-caterpillar insects.

> **Note:** Tent caterpillars follow natural multi-year population cycles. One or two seasons of modest defoliation are tolerated easily by healthy, well-managed trees.

8.4 Mouse and rabbit damage — trunk protection

Meadow voles and cottontail rabbits feed on bark at ground level and under snow. If they chew the bark right around the trunk the tree is **girdled** — its supply lines between root and canopy are severed — and it will die. This is the most common cause of young-tree loss on sites like this one, and the damage is invisible until the tree fails to leaf out. Inspect trunks in late October and again in late February or March.

- **White latex paint,** diluted 50:50 with water, applied from ground level to the first scaffold branches each autumn. It deters gnawing and protects against **sunscald**, the bark splitting that happens on the south-west side of a trunk when late-winter sun warms it by day and it refreezes at night. Use only diluted interior latex — undiluted or oil-based paint prevents the bark from breathing. **Mixing a little clean sand into the paint** makes the coating gritty and far less palatable to gnawing rodents.
- **Hardware cloth cylinders** (6 mm galvanised mesh, 60 to 90 cm tall) give the most durable barrier where pressure is severe. Leave a gap between the mesh and the trunk, and seat it below the soil surface.
- The dwarf apples have thin bark throughout their lives — hardware cloth is strongly recommended for the Cortland dwarfs and the trees on B.9 and B.10.
- **Burlap wrap.** Wrapping the trunk in burlap each autumn gives a simple physical barrier against gnawing and against sunscald, and it breathes in a way that plastic does not. Fit it in the fall and take it off, or at least loosen it, in spring.
- **Nothing wrapped round a trunk may be left to constrict it.** If burlap is left on through the growing season, check that it is not tightening as the trunk expands. The same applies to grafting tape, tree ties, spiral guards and hardware cloth — inspect every wrap twice a season and loosen or cut anything that has begun to bite in. A wrap that girdles a trunk does exactly the damage it was fitted to prevent.
- **Do not bury the stems.** Keep mulch from piling against the trunks, and pull it back before winter.

8.5 Recovering a girdled tree

A girdled tree is not necessarily a lost planting position. The roots below the damage are usually untouched and still hold a season's stored reserves, and if the top is taken off they will often push new shoots and rebuild a crown within a couple of years. That is much quicker than starting again with a new whip, and it keeps the row intact.

- **Cut the trunk off below the damage, and do it while the tree is still dormant** — winter or early spring, before it attempts to leaf out. The timing is the whole of it. A girdled tree allowed to break bud in spring spends its stored reserves pushing growth that the severed bark can no longer supply water to, and by the time it fails the roots have nothing left to re-flush with. Cut while dormant and those reserves go straight into new shoots instead. Handled this way the root system survives and the position stays in production without replanting.
- **Look at where the damage sits relative to the graft union** — the swelling low on the trunk where the variety was joined to the rootstock. If there is sound bark and live buds above the union, cut back to just above it and the regrowth will be your named variety. Select the strongest shoot as the new leader and remove the rest.
- **If the girdle runs to or below the graft union, everything that regrows is rootstock.** It will grow away strongly and it will carry the rootstock's vigour, size control and disease resistance — but its fruit will be the rootstock's, which is not what was planted.
- **Graft your chosen variety onto that new growth.** Let the shoots run for a season to thicken, then graft the following April as set out in Section 12. This gives the best of both: the established root system beneath, with its growth and disease resistance intact, and the fruiting variety you actually want above. It is also a free choice of variety — a chance to bring in a heritage sort, or a pollinizer where the block could use one.
- Reduce to a single strong leader once the graft has taken, and fit a trunk guard to the new stem from the outset. A tree that has been girdled once is standing exactly where the voles found it convenient the first time.

8.6 Japanese beetle

Japanese beetle (*Popillia japonica*) is an increasing problem in southern Ontario. Adults are active from late June through August, skeletonising leaves and attacking ripening fruit.

- **Hand-picking is the most effective immediate control** for an orchard this size. Pick in the early morning while the beetles are slow and drop them into soapy water, every one to two days during peak activity.
- **Neem oil** at the first sign of activity acts as a feeding deterrent. Begin early — neem is much less effective once a large population has established.
- **Grub treatment:** beneficial nematodes (*Heterorhabditis bacteriophora*) applied to grass in late summer or early autumn kill grubs before they emerge. Apply when soil is above 15 °C and keep it moist.

> **Caution:** Do not use Japanese beetle pheromone traps within or near the orchard. Research at the University of Kentucky has shown these traps attract more beetles from the surrounding area than they capture, increasing damage to nearby plants.

9. Pruning Guidelines

> **Note:** Pruning timing matters far less than pruning at all. Late fall, once the trees are fully dormant, and early spring before bud break are both good windows, and both suit every species in this orchard. Any other timing mentioned in this section is an ideal to aim for when the opportunity arises, not a rule.

9.1 General principles — keep trees short

The single most important long-term pruning decision is height management. Target a maximum of **2.5 to 3.0 m** for all trees in this orchard. That keeps fruit within reach from the ground or a single step, allows thorough spray coverage without powered equipment, and makes disease and pest monitoring practical. A low, open tree is a healthy, productive tree.

9.2 Training systems by species

- **Stone fruit** (apricot, plum, plumcot, cherry, nectarine, peach): train to the open-centre or vase system. Cut the central leader at 60 to 90 cm after planting — apricot, peach and nectarine at 60 cm, plum, plumcot and cherry at 90 cm — and select three to five **scaffolds** — the main structural limbs — radiating evenly. The **open centre** or vase form has no central stem at all, which lets light and air into the middle of the tree. The vase form admits light and airflow, which reduces brown rot. Always cut to an outward-facing bud.
- **Apple and pear:** train to a modified **central leader** — one dominant upright stem with three to four tiers of scaffolds arranged around it. Once the tree reaches 2.5 m, cut the leader back to hold height.
- **Dwarf apples** (Cortland dwarf, B.9 and B.10): maintain a simple central leader, remove vigorous upright competing shoots, and spur-prune each winter. A **spur** is a short stubby shoot that carries flower buds year after year, and pruning to them keeps fruiting concentrated and within reach.
- **Hazelnut:** grow as a multi-stemmed shrub or train to a single stem, whichever suits. Remove suckers annually and keep the centre open enough for airflow and inspection.
- **Pawpaw:** needs very little pruning. Remove dead or crossing wood only, and let the natural pyramidal form develop.

9.3 Annual pruning sequence

- **Step 1:** remove the four Ds first — Dead, Diseased, Damaged and Crossing or rubbing branches.
- **Step 2:** remove vigorous upright water sprouts growing into the centre of the tree, and any suckers from below the graft union.
- **Step 3:** head back any branch above your target height to a lower lateral.
- **Step 4:** thin for light. A rough guide — you should be able to throw a hat through the tree without it catching.
- **Step 5, nectarine and peach:** fruit is carried only on wood that grew the previous summer. After harvest, prune two-year-old fruiting wood back to a strong new shoot. This annual renewal is essential to keep the six Harkos productive.

9.4 Tools, timing and technique

- Use sharp bypass pruners. Cut 6 to 8 mm above an outward-facing bud at a slight angle, and leave no long stubs.
- Remove and dispose of all prunings. Diseased wood must be burned or bagged, not composted.
- **Timing — fall or early spring, and that is fine.** In practice the pruning gets done when there is time for it, and for most people that means **late fall, once the leaves are off and the trees are fully dormant, or early spring before bud break**. Both windows work well for every species in this orchard. Do not put pruning off waiting for a better moment — a tree pruned in the ordinary window every year is in far better shape than one pruned to the perfect textbook date every third year.
- **Prune before spraying** so there is less material to cover and the spray goes on the wood you are keeping.
- **Refinements, if the opportunity arises.** These are ideals to aim for rather than requirements. Stone fruit heals fastest pruned late in the dormant window, closer to bud break. Cherries are traditionally pruned in summer, July to August, when bacterial canker is least likely to enter the cuts. Apple and pear are best not pruned in warm humid weather, when fire blight moves into fresh wounds. Follow these where you can; where you cannot, prune in the fall or early spring window and keep the cuts clean and the tools sharp.

9.5 Berry planting

- **Cane fruit** (raspberry, blackberry): a cane grows one year as a **primocane**, fruits the next year as a **floricane**, then dies. Remove spent floricanes at ground level immediately after harvest, and thin the new ones. This single job keeps the row productive and open.
- **Bush fruit** (red currant, gooseberry, elderberry): renewal-prune in late winter, removing the oldest third of the stems at the base each year.
- **Goji** fruits on the current season’s growth, so prune it hard in late winter.

- **Saskatoon** needs only light thinning of the oldest stems every few years.
- **Strawberry** is **renovated** rather than pruned — shear the foliage off after fruiting and thin the runners, which resets the bed for the following year.
- **Staking: install only as necessary.** The dwarf apples, including the three on M27, are standing unaided at present and are best left that way — an unstaked tree builds a stronger, more tapered trunk. Check them annually for any sign that support has become necessary (Section 11.3). The Honeycrisp on G.969 is worth staking through its first several years given the exposure. Trees on vigorous roots need no support.

10. Orchard Floor — Mulch, Weeds and Mowing

The ground under and between the trees needs as much routine attention as the trees themselves. Grass and weed competition is the biggest brake on establishment growth, and long vegetation is an open invitation to the voles that girdle trunks under snow.

10.1 Mulch beds

Maintain a mulch ring about 1 m across and 8 to 10 cm deep around each tree, of wood chip or bark. It conserves moisture, suppresses the competition that most limits young-tree growth, and evens out soil temperature.

- **Do not bury the stems.** Keep the mulch clear of the bark all the way round, and pull it back from the trunks before winter.
- **Pull weeds out of the mulch beds every few weeks** through the growing season. Weeds that establish in a mulch ring are competing directly with the tree for exactly the water and nitrogen the mulch was laid down to conserve, and they are far easier to pull from loose mulch than from packed ground. Hand-pulling every two or three weeks keeps the job to a few minutes a bed.
- Top the mulch up each autumn as it breaks down, keeping the depth at 8 to 10 cm.

10.2 Mowing

Mow weekly through the growing season. Keeping the vegetation short between and around the rows is a pest-control measure as much as a tidiness one: voles and mice need cover, and short grass simply does not give them any. An orchard mown weekly is far less attractive to rodents than one mown occasionally, and the difference shows up as trunk damage found the following spring.

- Mow along the rows rather than across them, using the open ground at each end as the headland for turning.
- Mow shortest of all going into winter, so there is no standing cover left under the snow.
- Keep the mower and trimmer away from the trunks. A string trimmer will girdle a young tree as effectively as a vole; the mulch ring exists partly to keep machinery at a distance.

11. Rootstocks — What They Are and Why These Were Chosen

Almost every fruit tree in this orchard is two plants joined together. The **rootstock** is the root system; the **scion** is the fruiting variety grafted on top of it. The join shows as a swelling low on the trunk, the **graft union** — and anything sprouting from below that swelling belongs to the rootstock, not to the variety you paid for, so it should always be removed.

This matters more than most people expect. **The rootstock, not the variety, decides how big the tree gets, how soon it fruits, whether it needs a stake for life, how much water and feeding it demands, and how well it resists several root diseases.** Two Honeycrisp trees on different rootstocks can differ by a factor of three in mature size. That is why rootstock is recorded tree by tree in the planting record, and why it is worth understanding the ones used here.

11.1 A quick scale

Apple rootstocks are graded from extremely dwarfing to full-size seedling. Roughly, in increasing order of size: M27, then B.9 and B.10, then M9, M26, G.969, and finally seedling. A dwarf tree fruits sooner and is picked from the ground; a larger tree takes longer to bear, lives longer, and needs a ladder. Neither is better — they suit different jobs, and this orchard uses both.

11.2 The rootstocks used here

Rootstock	Trees	Size	Why it was chosen, and what it needs
M27	Jonagold (Row 1, Tree 10); Gravenstein (Row 4, Trees 6 and 10)	Extremely dwarfing — the smallest apple rootstock there is, about 1.5–2 m, roughly a third the size of a tree on M9	**This is a deliberate and rather expert choice.** All three M27 trees here are **triploids** — Jonagold and Gravenstein both carry three sets of chromosomes, which makes them naturally vigorous, large-fruited, big trees. M27 is the classic remedy: it is used specifically to cut the vigour of triploid varieties down to a manageable size. It is also very precocious, so these will fruit young. **Support: monitor rather than assume.** M27 is often described as needing a permanent stake, because its roots and stem can be too weak to carry the tree. **These three are standing solidly on their own at present and are deliberately unstaked.** That is fine while it lasts, and an unstaked tree builds better trunk taper. But the load grows as the canopy fills out and the first heavy crops arrive, so check them each year — see 11.3 for what to look for — and be ready to stake if they begin to lean or labour. Their other needs are not optional: shallow roots and very poor tolerance of dry soil, waterlogging or poor ground, and no ability to compete with grass, so keep the weed-free mulch circle maintained and give them priority on water and feeding. Also highly susceptible to fire blight, and often shorter-lived than other stocks.
B.10	Belmac (Row 4, Tree 5)	Dwarfing, about 2–2.5 m	A Budagovsky rootstock from Russia, bred for cold climates — which is exactly why it suits this site. Very winter-hardy roots, precocious, and resistant to fire blight and crown rot. Needs permanent support and a clear, weed-free root zone like any dwarf. Paired with Belmac, itself a disease-resistant Quebec variety, this is one of the lowest-maintenance trees in the orchard.
Dwarf (unspecified)	Cortland ×2 (Row 7, Trees 2 and 4); Ambrosia (Row 7, Tree 8); Williams' Pride (Row 7, Tree 9); Van cherry (Row 3, Tree 6)	Dwarfing	Recorded simply as dwarf. Almost certainly B.9 or a similar dwarfing stock for the apples. Treat all of them as needing permanent support, a clear weed-free circle and priority on water. If the nursery invoices are to hand it is worth adding the exact stock to the planting record, since it changes the spacing these trees will eventually need.
G.969	Honeycrisp (Row 3, Tree 7 — nursery tag 2025-G969)	Semi-dwarf, roughly 50–60% of a seedling tree	A Geneva rootstock, and the most disease-resistant stock in this orchard: it resists fire blight, crown rot and woolly apple aphid at the roots, and throws few suckers or burr knots. That resistance is in the roots only — the Honeycrisp scion above is still fire-blight-susceptible, so blossom-blight practice does not change. More precocious than its vigour suggests. Stake it through its first several years given the exposure here, then it can stand alone. It will outgrow the dwarf apples around it, so prune it to keep it in scale with the cherry row it sits in.
OH×F.97	Honey Sweet pear (Row 4, Tree 8); Douglas pear (Row 6, Tree 5)	Semi-dwarf to semi-vigorous	An Old Home × Farmingdale selection, and the standard pear rootstock in North America for one overriding reason: **fire blight resistance in the roots**. In an orchard containing three Bartletts, that matters. It is also winter-hardy and well-anchored, so these two do not need permanent staking. They will be among the larger trees in their rows, so prune them rather than letting them shade their neighbours.
Semi-dwarf	Yellow Egg plum (Row 4, Tree 9)	Semi-dwarf	Recorded as semi-dwarf. Plum rootstocks in this class — Marianna, St Julian and similar — give a tree of moderate size that is well anchored and does not need permanent support.

Rootstock	Trees	Size	Why it was chosen, and what it needs
Seedling / standard	The remaining trees, where no rootstock was recorded	Full size unless the variety is naturally small	Most of the 70 trees have no recorded rootstock. Where nursery invoices survive, adding them to the planting record is the single most useful piece of record-keeping still outstanding — it determines how much room each tree will eventually need and whether it will ever require support.

11.3 What this means in practice

- **Currently unstaked, and standing well:** the three M27 trees (Jonagold at Row 1 Tree 10, and the two Gravensteins at Row 4 Trees 6 and 10). Leave them as they are for now. **Inspect them once a year** — late winter, while pruning, is the natural time — and stake if any of these appear: a lean that was not there last season; the trunk flexing at or just below the graft union rather than through the whole stem; roots lifting or the soil cracking on one side after wind; or the tree bending under a heavy crop. The first substantial fruit load is the usual trigger, so watch them most closely in the first cropping years. A stake fitted before a tree leans is easy; one fitted afterwards rarely straightens it.
- **Watch similarly:** the B.10 Belmac and the four recorded dwarf apples in Row 7 and Row 3. Same annual check, same triggers.
- **Stake for a few years, then release:** the G.969 Honeycrisp, given the exposure at its position.
- **No permanent staking needed:** the two OH×F.97 pears, the semi-dwarf plum, and anything on seedling roots — these build a stronger trunk taper if left to move in the wind.
- **Water and weed priority:** the M27 and dwarf trees first, always. Their root systems are small and shallow, they cannot forage, and grass growing to the trunk will measurably slow them. This is what the 60 cm weed-free circle and the mulch ring are for.
- **Fire blight watch:** M27 is highly susceptible in the rootstock itself, so watch the Jonagold and the two Gravensteins for strikes running down into the stock, as well as the Bartletts and Honeycrisps above ground.
- **Remove suckers** from below the graft union on every tree, every year. On a dwarf tree especially, rootstock suckers steal growth the scion cannot spare.

12. Grafting — Adding Varieties to Existing Trees

Grafting a new variety onto an established tree is the fastest way to widen what this orchard grows. Working onto a tree that is already in the ground, with a root system and a framework of branches behind it, gives fruit years sooner than planting a new whip would, and it costs nothing but a stick of wood and an afternoon. It is the traditional method of keeping heritage varieties alive, and a single mature tree can carry several.

Three good reasons to graft here: to add more of a favourite fruit without giving up a planting position; to bring in heritage varieties that are no longer sold as nursery trees; and to use mature scion wood, which comes into bearing sooner than juvenile wood.

12.1 Timing

Graft in the first week of April. The rootstock tree should be just waking up while the scion wood is still fully dormant, and that is the window where the two knit together most reliably. Collect scion wood earlier in the winter while it is dormant and keep it cold until the day.

12.2 Wrapping the graft

- Bind the union with **grafting tape**, and carry the wrapping over the **whole scion**, not just the join. The tape holds the cut surfaces in firm contact and, just as important, stops the scion drying out before it can draw water from its new host.
- **Cover the wrapped graft and scion with tinfoil.** The foil shades the union and the buds, keeps the temperature even, and prevents the scion being forced into growth by direct sun before the union has knitted.

12.3 Aftercare — first week of May

Check the grafts in the **first week of May**. Where buds are swelling and pushing, the union has taken and the wrapping now needs opening in stages.

- **Unwrap the tape from the expanding buds** so they are not constricted, leaving the tape on the union itself.
- **Open the tinfoil gradually.** Expose the new growth to sunlight a little at a time rather than all at once. Keep using the foil to shade the buds themselves, and do not expose them fully until the leaves are properly expanding — soft new growth that has developed under cover will scorch if it meets full sun in a single step.
- Once the leaves are expanding and hardened, the foil can come off entirely. Watch the tape on the union through the season and slit it before it begins to bite into the growing wood.

> **Caution:** Anything wrapped round a stem — grafting tape, burlap, ties, guards — must never be left to constrict it. Wood expands through the season; a wrap that was comfortable in April can girdle the same stem by August. Check every wrap on the property twice a season and loosen or slit anything that has tightened.

13. Watering & Establishment

Newly planted trees have a small, disturbed root system and cannot yet reach moisture deeper in the soil profile. Consistent watering through the first one to two growing seasons is the single biggest factor in survival and strong establishment. The goal is steady, even soil moisture — never bone-dry, never waterlogged.

13.1 Routine schedule

- Water **every 3 days**. By hand, that is roughly **20 seconds per tree** from the hose at a steady flow, delivered slowly at the base so it soaks in rather than running off. On the drip system it is about **1 hour 20 minutes** per session — see Section 14.1.
- Increase frequency and duration in hot, dry or windy spells — during heat waves water every one to two days and lengthen each watering. Fast-draining spots dry sooner and need more.
- Water in the early morning where possible, at the root zone rather than over the canopy.

- Check before watering in wet weather. If the top 5 to 8 cm of soil is still moist, skip that cycle.
- **Check the soil one day after watering.** Dig down beside a mulch ring with a trowel or push a finger in to 10 cm. The soil should be evenly damp, holding together when squeezed but not glistening or muddy. If it is soaking wet a full day later, the volume or the frequency is too high.
- **If you find saturated soil, stop watering for the time being.** Keep checking daily until it has drained to evenly damp, then start again at a reduced volume or a longer interval. Roots in waterlogged soil cannot breathe, and a tree drowned slowly looks much like a tree dying of drought — wilting, yellowing leaves — which makes it easy to respond by watering more and compound the problem.
- Adjust volume and frequency together as the season and the soil dictate rather than holding to a fixed schedule. Free-draining ground will take the full rate; heavier or low-lying spots will need less, and the day-after check is what tells you which you have.

13.2 Technique and tapering off

- Maintain a shallow basin or mulch ring around each trunk to hold water over the root ball, keeping the mulch 8 to 10 cm clear of the trunk itself.
- The three pawpaws are the most moisture-sensitive trees in the orchard — do not let them dry out in year one. The dwarf apples, and above all the three on M27, are next: their root systems are small and shallow and cannot forage for water.
- Through late summer, begin tapering so growth hardens off before winter. Stop routine watering once the trees are dormant, apart from a deep soak before freeze-up if autumn is dry.

13.3 Established trees — caliper-based rate and rainfall

Once established, a reliable rule of thumb is **4 litres of water per centimetre of trunk caliper, per week, per tree**. **Caliper** is simply the diameter of the trunk, measured 15 cm above the ground with a tape or callipers. A 3 cm-caliper tree therefore needs about 12 L a week and a 5 cm tree about 20 L. The drip system here runs at 8 minutes per litre, which converts that rule directly into a timer setting — see Section 14.1.

Rainfall alone is usually not enough to wet the rooting zone for a full week. It takes at least about **30 mm of cumulative rain over a 7-day period** to recharge the root zone properly, and in this area that typically happens only **three or four times across a whole growing season**. Plan to supplement for the rest of the season rather than counting on rain. A rain gauge on the fence, read and recorded, is worth having.

> **Caution:** Over-watering is as damaging as drought — constantly saturated soil suffocates roots and invites root rot. Let the surface begin to dry between waterings; the 3-day rhythm above assumes free-draining soil and should be eased on heavy or poorly drained ground.

14. Irrigation System

A buried half-inch line serves the orchard from a **tap on the north fence**, close to the north-western end of the planted block. From there the header runs south down the western end of the rows, feeding a lateral along each of the seven rows, each set approximately **1 m to the south of its row centreline**. Beyond Row 7 the line continues out to the **south fence**, turns to run about 36 m north-east along it, roughly 3 m inside and parallel to the boundary, then turns again at the east corner for a final **12 m leg north up the east fence**. The berry hedge follows that south and east run, so the fence line is irrigated on the same schedule as the orchard.

This layout removes the labour of hand-watering, delivers water evenly to each root zone, and is easily calibrated to the caliper-based weekly volume above — particularly valuable through fruit sizing and in hot, dry stretches.

- **Mark the line.** Record the 1 m southerly offset from each row centreline so the line is never cut by a fence post or a planting hole.
- **Blow out or drain the system** before the first hard frost.

- **Check emitter output at the far end of each lateral** early each season. Pressure loss shows first at the eastern end of the longest run, and an under-watered end tree can look much like a diseased one.
- **Walk the system regularly through the season** — weekly is not too often, and it fits naturally into the same walk as the pest scouting. A buried line fails silently: the trees look fine for a fortnight and then all of them are in trouble at once. Look for wet patches or sinkholes over the buried run, which mean a leak; dry, dusty ground at an emitter that should be running; and unexplained pressure loss along a lateral.
- **Confirm the water is actually on.** A tap knocked shut, a timer that has lost its program after a power interruption, or a valve closed for another job and never reopened are the commonest causes of an orchard going dry, and all three are invisible unless someone looks.
- **Check the nozzles are intact and in place.** Emitters get knocked off by mowers and animals, split with frost, or silt up until they barely trickle. Count them against the trees as you walk, and carry a few spares.
- **Run the rows in shifts if the tap cannot keep up.** With every emitter open at once the system draws roughly 8.75 L per minute; if pressure drops at the far end, water Rows 1 to 4 and then Rows 5 to 7 as two separate cycles.

14.1 Flow rate and run times

The system has been measured at **8 minutes per litre** at the emitter — that is, each tree receives about 7.5 litres an hour. This is a normal drip rate and the slowness is the point: water applied gently over hours soaks down into the root zone instead of running off the surface or spreading sideways across the grass.

That single figure turns the caliper rule in Section 13.3 into a timer setting. Four litres per centimetre of caliper per week, at 8 minutes a litre, gives a rule worth writing on the controller:

> **Note: Weekly run time in minutes = 32 × trunk caliper in centimetres.** A 3 cm tree needs 1 hour 36 minutes a week; a 5 cm tree needs 2 hours 40 minutes.

Split the weekly total across two or three sessions rather than delivering it in one long run — the soil holds it better and the roots are never sitting saturated. During establishment, when the target is roughly 10 litres per tree every three days, that is about **1 hour 20 minutes per session**.

Trunk caliper	Water per week	Run time per week	Split over 2 sessions
1 cm	4 L	32 min	16 min each
2 cm	8 L	1 h 04 min	32 min each
3 cm	12 L	1 h 36 min	48 min each
4 cm	16 L	2 h 08 min	1 h 04 min each
5 cm	20 L	2 h 40 min	1 h 20 min each
6 cm	24 L	3 h 12 min	1 h 36 min each
8 cm	32 L	4 h 16 min	2 h 08 min each
10 cm	40 L	5 h 20 min	2 h 40 min each
12 cm	48 L	6 h 24 min	3 h 12 min each

Times assume one emitter per tree. Where a tree carries two emitters, halve the run time. Check the rate again each spring by timing how long an emitter takes to fill a one-litre jug — emitters silt up gradually, and a rate that has drifted to 12 minutes a litre means the trees are getting two-thirds of what the timer says.

15. Fertilizing

Use a **granular slow-release fertilizer formulated for deciduous trees** — a tree-and-shrub or fruit-tree blend. Slow-release granules feed gradually over the season, avoiding the flush of soft, disease- and frost-prone growth that quick-release products can cause. Apply strictly per the product label, which sets the rate by tree size or trunk diameter.

- **Timing and frequency:** follow the directions on the product. Begin in early spring as growth starts, around bud swell once the ground has thawed. Because this is a granular slow-release fertilizer, **the label may call for reapplying every few weeks** through the growing season — apply at whatever interval and rate the product specifies rather than assuming a single spring dose is enough.
- **Placement:** spread the granules evenly over the root zone, broadcasting out to and just beyond the **drip line** — the circle on the ground directly beneath the outer tips of the branches, which is where most of the feeding roots sit. Keep granules off the trunk and away from the root flare to avoid burn.
- **Water it in** after application so the granules begin to release into the root zone.
- All trees here are deciduous, so the same blend suits the apples, pears, stone fruit, hazelnuts and pawpaws alike.
- **The dwarf trees need this most.** Trees on M27 and the other dwarfing stocks have small, shallow root systems that cannot forage, so they depend on what is applied within their own mulch circle.
- **Do not fertilize after midsummer.** Late nitrogen pushes soft growth that will not harden before winter.

A soil test every three to four years is worth more than any fertilizer program run on assumption, particularly for pH, which governs whether the nutrients present are available to the trees at all.

16. Complete Tree Inventory

Seventy trees as planted. Row 1 is the northern row; Tree 1 is the western end of each row. Rows are 5.69 m apart and trees 2.98 m apart within the row.

Row.Tree	Variety	Species	Stock
1.1	Harko	Nectarine	—
1.2	Harko	Nectarine	—
1.3	Harko	Nectarine	—
1.4	Gala	Apple	—
1.5	Italian	Plum	—
1.6	Italian	Plum	—
1.7	Harko	Nectarine	—
1.8	Harko	Nectarine	—
1.9	Harko	Nectarine	—
1.10	Jonagold	Apple	M27
2.1	Moorpark	Apricot	—
2.2	Fuji	Apple	—
2.3	Liberty	Apple	—
2.4	Moorpark	Apricot	—
2.5	Moorpark	Apricot	—
2.6	Puget Gold	Apricot	—
2.7	Tonda	Hazelnut	—
2.8	Gene	Hazelnut	—
2.9	Gene	Hazelnut	—
2.10	Tonda	Hazelnut	—
3.1	Bing	Cherry	—
3.2	Bing	Cherry	—
3.3	Sam	Cherry	—
3.4	Bing	Cherry	—
3.5	Montmorency	Cherry	—
3.6	Van	Cherry	dwarf
3.7	Honeycrisp	Apple	G.969
3.8	Stella	Cherry	—
3.9	Fuji	Apple	—
3.10	Glacier	Cherry	—
4.1	Northern Sunset	Plumcot	—
4.2	Northern Sunset	Plumcot	—
4.3	Shiro Gold	Plum	—
4.4	Northern Sunset	Plumcot	—
4.5	Belmac	Apple	B.10
4.6	Gravenstein	Apple	M27
4.7	Northern Sunset	Plumcot	—
4.8	Honey Sweet	Pear	OHxF.97
4.9	Yellow Egg	Plum	semi-dwarf
4.10	Gravenstein	Apple	M27
5.1	Spartan	Apple	—
5.2	Flemish Beauty	Pear	—
5.3	Gala	Apple	—
5.4	Intrepid	Peach	—
5.5	Dolgo	Crabapple	—
5.6	Liberty	Apple	—
5.7	Spartan	Apple	—
5.8	Pawpaw	Pawpaw	—
5.9	Pawpaw	Pawpaw	—
5.10	Pawpaw	Pawpaw	—
6.1	Korean Giant	Asian pear	—
6.2	Bartlett	Pear	—
6.3	Bartlett	Pear	—
6.4	Bartlett	Pear	—
6.5	Douglas	Pear	OHxF.97
6.6	Bosc	Pear	—
6.7	Bosc	Pear	—
6.8	Red Clapp	Pear	—
6.9	Bosc	Pear	—
6.10	Bosc	Pear	—
7.1	Gala	Apple	—
7.2	Cortland	Apple	—
7.3	Honeycrisp	Apple	—
7.4	Cortland	Apple	—
7.5	Gala	Apple	—
7.6	Honeycrisp	Apple	—
7.7	Gala	Apple	—
7.8	Ambrosia	Apple	dwarf
7.9	Williams' Pride	Apple	dwarf
7.10	Honeycrisp	Apple	—

17. Annual Management Calendar

> **Note: Dates are calibrated to this site, not to Niagara.** Spring runs about two weeks later here and harvest three to four weeks later than the Ontario fruit-growing advice you will find published elsewhere. Treat them as a typical year and let the trees themselves set the actual timing — see Section 1.3.

> **Note: This calendar is also built into the interactive map.** Open the map, enter any date — today, or one you are planning ahead for — and it lists the tasks due in the following two weeks, month or two months, each with the section of this guide that explains it. If you are ever unsure what the orchard needs, that is the quickest place to look.

Stone fruit is managed first each season, following the bloom sequence. Confirm all spray products against current OMAFRA domestic-use recommendations each year, and always read the product label before use. Establishment watering — about every 3 days, roughly 20 seconds per tree, more in heat or drought — runs through the growing season in years one and two, and slow-release granular fertilizer is applied per the product label. Both are folded into the months below.

Month	Key tasks	Products / methods	Priority species
Oct – Nov	Apply lime sulphur after leaf drop for peach leaf curl. Remove all mummified fruit from trees and ground. Late apple and pear harvest. Dormant pruning begins. Inspect trunks and apply diluted white latex paint with a little sand mixed in. Do not bury the stems. Begin tent caterpillar egg-mass inspection.	Lime sulphur (NOT on apricot). Diluted white latex paint 50:50 with sand. Manual egg-mass removal.	Nectarine, peach, apple, pear. All trunks.
Nov – Feb	Complete dormant pruning — the late-fall window is equally good. Check the dwarf and M27 trees for any new lean or trunk flex and stake if needed (Section 11.3). Inspect trunks for winter rodent damage and cut any girdled tree off below the damage now, while it is still dormant, so the roots re-flush (Section 8.5). Inspect plums, plumcots and cherries for black knot and prune 20 cm below galls. Inspect the four hazelnuts for Eastern filbert blight and cut affected wood well back. Continue egg-mass inspection. Order supplies: stakes, frost cloth, dormant kit, traps and lures. Renewal-prune the currants, gooseberries and elderberry; cut the goji back hard.	Manual pruning and scouting.	All. Hazelnut, plum, plumcot, cherry.
Late Mar – late Apr	Apply the Green Earth Dormant Spray Kit to all trees EXCEPT the apricots, which receive oil only. Apply before any green shows at the bud tips. Second lime-sulphur spray on nectarine and peach if leaf curl was bad the prior year. Watch the apricots for frost through their mid-to-late April bloom and deploy cloth below –2 °C — last frost here averages mid-May.	Green Earth kit: 120 ml lime sulphur + 60 ml oil per 3 L. Apricot: 60 ml oil only per 3 L. Frost cloth.	All trees. Apricot oil-only.
Late Apr – May	**Graft in the first week of April** — wrap scion and union with tape and cover with foil (Section 12). Copper at silver or green tip for fire-blight suppression on pear and apple. First brown-rot fungicide on apricot and plum at 10% bloom. Hand-pollinate the pawpaws through bloom, which peaks around the third week of May here. Watch for the first tent caterpillars. Apply slow-release granular fertilizer per label. Begin establishment watering. Commission the irrigation system and check emitter output at the end of each lateral.	Copper fungicide. Btk if larvae active. Slow-release granular fertilizer. Soft brush for pawpaw. Grafting tape and foil.	Pear, apple, apricot, plum, plumcot, pawpaw, Saskatoon.
Mid May – mid Jun	**Check the grafts in the first week of May** — unwrap tape from expanding buds and open the foil gradually. Brown-rot sprays at pink bud and petal fall on all stone fruit. Fire-blight monitoring on the three Bartletts. Thin apricot, peach and nectarine fruitlets to one per 15 cm after petal fall. Wettable sulphur for apple scab every 7–14 days in wet weather. Hang codling moth traps at petal fall and apple maggot traps late in the month. Kaolin clay at petal fall for plum curculio.	Wettable sulphur (NOT on apricot). Copper. Kaolin clay. Pheromone and sticky-sphere traps.	All stone fruit. Apple, pear.
Mid Jun – Jul	First codling moth spray when moths are caught — typically late June here. Go by the traps, not the date. Continue scab sprays if wet. **Mow weekly** and pull weeds from the mulch beds. Walk the irrigation line for leaks, closed taps and missing emitters, and check the soil a day after watering. Thin apple fruitlets to one per cluster. Monitor for plum curculio. Begin hand-picking Japanese beetles. Water every 3 days, more in heat — the pawpaws must not dry out.	Btk or spinosad. Wettable sulphur. Kaolin clay. Neem oil. Trowel for soil checks.	Apple, pear, plum, Saskatoon, all fruit.

Month	Key tasks	Products / methods	Priority species
Mid Jul – Aug	Continue weekly mowing and the fortnightly weed-pull in the mulch beds. Check tree ties, guards and any grafting tape for constriction. Second codling moth spray about 21 days after the first. Monitor apple maggot traps. Brown-rot pre-harvest spray on nectarine and peach. Summer-prune the cherries if there is time — it reduces bacterial canker, though the dormant window works too. Continue hand-picking Japanese beetles. Apply beneficial nematodes for grub control. Continue watering through heat. Harvest strawberries, raspberries and currants; cut spent raspberry floricanes to the ground as each planting finishes.	Spinosad. Insecticidal soap or summer oil. Kaolin clay. Beneficial nematodes.	Apple, pear, nectarine, peach, cherry.
Late Aug – Sep	Harvest nectarines, the Intrepid peach and the cherries — late August into September on this site, three to four weeks behind Niagara. Second codling moth generation peak — monitor traps. Japanese beetles wind down by mid-month. Begin tapering watering in late August so growth hardens before winter. No fertilizer after midsummer. Harvest blackberries and elderberries; cut spent blackberry canes after picking.	Spinosad if needed. Reduce watering.	Nectarine, peach, cherry, apple.
Mid Sep – Oct	Main apple harvest, Honeycrisp into October here. Pear harvest — Bartlett, Bosc, Douglas, Red Clapp. Pawpaw harvest late September into October, when the fruit softens and drops — it will not ripen off the tree. Hazelnut harvest as husks brown. Brown-rot clean-up — remove all mummified fruit. Apply lime sulphur or copper as leaf drop begins for peach leaf curl. Goji and autumn strawberries continue cropping. Renovate the strawberry bed.	Lime sulphur or copper (NOT lime sulphur on apricot).	Apple, pear, pawpaw, hazelnut, nectarine.
Before hard frost	Drain or blow out the irrigation line. Collect all fallen fruit. Top up mulch around all trees, kept clear of the stems. Pull mulch back from trunks and mow shortest of all to remove vole cover. **Wrap trunks in burlap** as a physical barrier for the winter. Confirm trunk guards are seated below grade and above the expected snow line.	Irrigation shutdown. Burlap wrap. White latex paint. Hardware cloth check.	All trees.

18. Variety Notes

Every variety planted here was chosen for a reason — flavour, hardiness, disease resistance, or the job it does pollinating its neighbours. This section sets out what each one is, where it sits, what it needs by way of a pollen partner, and what to expect from it. It is written to be browsed rather than read straight through: look up a variety when you are standing in front of it and wondering what it is.

18.1 Trees

Apple

Variety	Where	Pollination	Notes
Ambrosia	Row 7, Tree 8 (dwarf)	Needs a pollen partner	A British Columbia chance seedling from the 1980s and now one of Canada's most planted apples. Very sweet, low acid, aromatic, with cream-coloured flesh that browns unusually slowly — which makes it the best apple here for lunchboxes and fruit plates. Ripens mid-season. Bruises easily, so pick and handle gently.
Belmac	Row 4, Tree 5 (B.10 dwarf)	Needs a pollen partner	A Quebec-bred apple developed as an improvement on McIntosh, from a complex cross including Melba and Spartan. Its great merit is disease resistance — it is one of the standard scab-resistant varieties recommended for cold Canadian gardens, and very hardy. McIntosh-type flavour, good fresh and for sauce. On B.10 it will stay small and crop early.
Cortland	Row 7, Trees 2 and 4 (dwarf)	Needs a pollen partner	An 1898 New York cross of Ben Davis and McIntosh, and a long-standing favourite in Ontario. Snow-white flesh that resists browning better than most, which makes it the classic salad and pie apple. Tender skin, sweet-tart, ripens late September. Does not store as long as a Fuji or Granny Smith — use it within a couple of months.
Fuji	Row 2, Tree 2 and Row 3, Tree 9	Needs a pollen partner	A Japanese cross of Red Delicious and Ralls Janet, introduced in the 1960s. Dense, very sweet, crisp, and the best keeper in this orchard — it will hold in a cold room for months and actually improves for a few weeks after picking. Ripens late, so it wants the warmest position and a long autumn. Needs thinning or it will biennial-bear.
Gala	Row 1 Tree 4, Row 5 Tree 3, Row 7 Trees 1, 5 and 7	Needs a pollen partner	A New Zealand cross of Kidd's Orange Red and Golden Delicious. Mild, sweet, crisp and thin-skinned, the apple children eat without complaint. Ripens early to mid-season, before most of the block. Susceptible to scab, so it belongs in the wettable-sulphur program. Five trees spread across three rows makes it the most widely distributed variety here.
Gravenstein	Row 4, Trees 6 and 10 (M27 dwarf)	**Triploid** — takes pollen, cannot give it	A very old variety, probably Danish, dating to the 1600s and prized above almost all others for cooking, sauce and cider. Tart, aromatic, juicy, and ripens early. Being triploid it produces little viable pollen, so it needs a diploid apple nearby and will not serve as a partner for anything else — Belmac sits immediately beside the Tree 6 plant and the Dolgo crabapple is one row away. On M27 these will be genuinely small trees needing permanent support. Does not keep; use it as it comes.
Honeycrisp	Row 3 Tree 7 (G.969), Row 7 Trees 3, 6 and 10	Needs a pollen partner	The University of Minnesota apple that changed what people expect from texture — explosively crisp, juicy, sweet with balancing acidity, and it holds that crispness in storage far better than it has any right to. Cold-hardy, which suits this site. Two cautions: it is susceptible to fire blight, and it is prone to bitter pit, a calcium-related disorder showing as sunken brown spots — which is worsened by heavy nitrogen, over-thinning and erratic watering. Steady moisture is the best prevention, which is what the irrigation system is for.

Variety	Where	Pollination	Notes
Jonagold	Row 1, Tree 10 (M27 dwarf)	**Triploid** — takes pollen, cannot give it	A 1943 New York cross of Golden Delicious and Jonathan, and among the best-flavoured apples grown — large, honeyed, aromatic, excellent fresh and for baking. Like Gravenstein it is triploid, so it receives pollen rather than supplying it; the Fuji and Liberty in Row 2 are within easy bee range. On M27 it will stay small, which is helpful because Jonagold is naturally a vigorous, large-fruited tree.
Liberty	Row 2 Tree 3 and Row 5 Tree 6	Needs a pollen partner	Bred at Geneva, New York and released in 1978, Liberty carries the Vf gene and is **fully resistant to apple scab** — it needs no scab spray at all. It also has good resistance to fire blight, cedar apple rust and mildew, which makes it about the lowest-maintenance apple you can plant in Ontario. McIntosh-like flavour that improves after a few weeks in storage, and it makes very good cider. Distributed through the block, the two Liberties lower the overall scab inoculum around the varieties that do need protection.
Spartan	Row 5, Trees 1 and 7	Needs a pollen partner	A 1936 introduction from Summerland, British Columbia — the first apple from a formal government breeding program in Canada. A McIntosh seedling with deep burgundy skin, bright white flesh, and a crisp snap McIntosh lacks. Excellent fresh, hardy, reliable, and keeps reasonably well. A good all-round choice for this climate.
Williams' Pride	Row 7, Tree 9 (dwarf)	Needs a pollen partner	A co-operative breeding release (the PRI program) and, like Liberty, **scab-resistant**, with useful resistance to fire blight and cedar apple rust as well. Its distinction is timing: it is one of the earliest apples of the season, dark red, sprightly and juicy, ready well before the main harvest. Does not store — eat it off the tree, which is the point of an early apple.

Crabapple

Variety	Where	Pollination	Notes
Dolgo	Row 5, Tree 5	Universal pollinizer	The keystone of the apple block. Dolgo is a Russian-derived crabapple grown as much for pollination as for fruit: it flowers long, heavily, and right through the middle of the apple bloom window, and is one of the best universal pollinizers available. Sited centrally, it serves the whole planting. The small crimson fruit are too tart to eat raw but make outstanding clear red jelly, and the spring blossom is beautiful. Keep it, prune it lightly, and do not judge it by the size of its fruit.

Pear

Variety	Where	Pollination	Notes
Bartlett	Row 6, Trees 2, 3 and 4	Cross-pollinates with all European pears	The world's most planted pear, known as Williams outside North America and dating to 1765 in England. Classic pear flavour, aromatic and buttery, superb fresh and the standard variety for canning. Picks green and ripens off the tree — left to ripen on the branch it goes gritty at the core. **The most fire-blight-susceptible planting in this orchard**: three together at Row 6 makes that block the first place to inspect each June.
Bosc	Row 6, Trees 6, 7, 9 and 10	Cross-pollinates with all European pears	A nineteenth-century Belgian or French variety, unmistakable for its long tapered neck and russeted cinnamon skin. Dense, spicy, honeyed flesh that holds its shape when cooked, which makes it the best poaching and baking pear here. Stores longer than Bartlett and ripens later. More fire-blight tolerant than Bartlett, though not immune.

Variety	Where	Pollination	Notes
Douglas	Row 6, Tree 5 (OH×F.97)	Cross-pollinates with all European pears	A hardy, productive European pear with better fire-blight resistance than Bartlett — which is why it sits in the middle of the pear row. Sweet, juicy, good fresh and for canning. On OH×F.97 the rootstock adds its own fire-blight resistance at the roots, making this one of the more robust trees in the block.
Flemish Beauty	Row 5, Tree 2	Cross-pollinates with all European pears	A Belgian variety from around 1810 and one of the hardiest European pears grown — it was the standard pear across cold parts of Ontario and Quebec for a century. Large, pale yellow with a red blush, richly sweet and aromatic. Susceptible to pear scab, so include it in the sulphur program in wet springs. Ripens early to mid-season.
Highland	Recorded in the nursery order; position to be confirmed	Cross-pollinates with all European pears	A 1974 release from a Bartlett × Comice cross, bred by Dr George Oberle. It is hardy in both warm and cold climates, and produces large, thin-skinned fruit of the highest quality — butter-smooth and highly flavoured when ripened at room temperature after a few weeks of cold storage. Harvest is very late, around the beginning of October, and it keeps up to two months. **Susceptible to both pear scab and fire blight**, so it belongs in the same monitoring routine as the Bartletts.
Honey Sweet	Row 4, Tree 8 (OH×F.97)	Cross-pollinates with all European pears	A sweet, aromatic dessert pear with good fire-blight resistance, which together with the OH×F.97 rootstock makes it a low-risk choice. It sits alone among the plumcots and plums in Row 4, so its pollen partners are the pears one and two rows south — comfortably within bee range.
Red Clapp	Row 6, Tree 8	Cross-pollinates with all European pears	A red-skinned sport of Clapp's Favourite, a Massachusetts variety from the 1860s. One of the earliest pears to ripen, sweet and fine-textured, with striking crimson skin. Like Bartlett it must be picked while still firm. Susceptible to fire blight.

Asian pear

Variety	Where	Pollination	Notes
Korean Giant	Row 6, Tree 1	Blooms earlier than the European pears	Also sold as Olympic or Dan Bae. A very large, round, russeted Asian pear — crisp and juicy like an apple rather than buttery like a European pear, and eaten firm straight off the tree. It is the best-keeping Asian pear, holding for months. Being an Asian pear it flowers a little earlier than its neighbours, so it sets on the shoulder of the European bloom; that is a characteristic of the planting rather than a fault. Fire-blight resistant.

Cherry

Variety	Where	Pollination	Notes
Bing	Row 3, Trees 1, 2 and 4	**Group III** — needs Sam or Van	The benchmark dark sweet cherry, bred in Oregon in 1875 and named after the orchard foreman Ah Bing. Very large, firm, deep mahogany, richly sweet. It is self-unfruitful and sits in S-allele Group III with Lambert and Napoleon, so it cannot be fertilised by those — but **Sam at Tree 3 and Van at Tree 6** are both established Bing pollinizers, and every Bing here is within two trees of one. Prone to rain-cracking near harvest.
Sam	Row 3, Tree 3	Pollinizer for Bing	A British Columbia variety from Summerland. Large, firm, black, good flavour, and notably **resistant to rain-cracking** — useful in an Ontario summer. Its main job in this row is pollination: Sam is a reliable partner for Bing and is deliberately placed among them.

Variety	Where	Pollination	Notes
Van	Row 3, Tree 6 (dwarf)	Pollinizer for Bing	Another Summerland introduction, and one of the most productive sweet cherries grown — it comes into bearing young and crops heavily. Dark red, firm, sweet. Like Sam it is a compatible Bing pollinizer, which is why the two flank the Bings from either side. On dwarf rootstock it will stay pickable from the ground.
Stella	Row 3, Tree 8	**Self-fertile**	The first self-fertile sweet cherry ever released, bred at Summerland in 1968, and the parent of most self-fertile cherries since. It needs no partner and contributes pollen broadly across the row. Large, dark, heart-shaped, sweet. Note that Stella is reported to perform poorly as a pollinizer specifically for Bing — which is why Sam and Van are the working partners in this row and Stella is a bonus rather than the plan.
Glacier	Row 3, Tree 10	**Self-fertile**	A modern self-fertile sweet cherry, early-ripening, large and dark with good crack resistance. Needs no partner and adds pollen to the row.

Tart cherry

Variety	Where	Pollination	Notes
Montmorency	Row 3, Tree 5	**Self-fertile**	The standard tart or sour cherry of North America, a French variety centuries old. Large, bright red, clear juice, far too sharp to eat raw and unmatched for pie, preserves and juice — this is the cherry worth planting if you bake. Self-fertile, so it crops alone, and hardier and later-blooming than the sweet cherries, which puts its flowers further from frost. It will not reliably cross with the sweet cherries because the bloom periods only sometimes overlap; it does not need to.

Apricot

Variety	Where	Pollination	Notes
Moorpark	Row 2, Trees 1, 4 and 5	Self-fertile	An English apricot known since the 1700s and still the standard for flavour — large, richly aromatic, orange flushed red, superb fresh, dried or in jam. It blooms very early, which is the whole difficulty with apricots in this climate: the flowers open before the frost risk has passed. Keep the frost cloth to hand. Self-fertile, and three together set more heavily than one would.
Puget Gold	Row 2, Tree 6	Self-fertile	A Washington State discovery, found as a seedling at the Anacortes ferry landing and named for its reliability in cool, wet springs. Its value here is timing — it is a **late-blooming** apricot bred precisely to dodge spring frost, and it sets and sizes fruit in conditions where other apricots fail. Large elongated golden fruit, freestone, sweet, ripening late August. Hardy to around –32 °C. Having it beside the three Moorparks spreads the apricot bloom across a wider window, which is exactly the right insurance on this site.

Nectarine

Variety	Where	Pollination	Notes
Harko	Row 1, Trees 1, 2, 3, 7, 8 and 9	**Self-fertile**	Bred at the Harrow Research Station in Ontario specifically for this climate, and the reason there are six of them — Harko is about as hardy and reliable as a nectarine gets here. Freestone, yellow-fleshed, red-blushed, sweet and juicy. Self-fertile, so all six crop independently. Remember that nectarines and peaches fruit only on wood grown the previous summer, so annual renewal pruning is essential (Section 9.3).

Peach

Variety	Where	Pollination	Notes
Intrepid	Row 5, Tree 4	**Self-fertile**	Chosen for hardiness above all. Intrepid is notable for **late bloom and exceptional bud hardiness**, which lets it dodge the spring frosts that catch other peaches — growers report crops in years when neighbouring peaches set nothing. Medium to large, red over yellow, freestone, good flavour. Same renewal pruning as the nectarines.

Plum

Variety	Where	Pollination	Notes
Italian	Row 1, Trees 5 and 6	European; partially self-fertile	The Italian prune plum, a European plum (*Prunus domestica*) and the classic drying and baking plum — deep purple-blue skin, amber freestone flesh, sweet and rich, holding its shape in a tart. Blooms later than Japanese plums, which helps with frost. Partially self-fertile and the pair here will cross-pollinate one another. **European and Japanese plums form separate pollination groups** and will not fertilise each other.
Yellow Egg	Row 4, Tree 9 (semi-dwarf)	European; partially self-fertile	An old European plum, large, oval and golden-yellow, traditionally grown for preserving and canning rather than fresh eating. Its pollination partners are the European plums and the plumcots in the same row.
Shiro Gold	Row 4, Tree 3	Japanese; needs a Japanese-type partner	A Japanese plum (*Prunus salicina*) — round, translucent yellow, very juicy and sweet, ripening early. Japanese plums bloom earlier than European ones and belong to a separate pollination group, so its working partners here are the Northern Sunset plumcots in the same row, which share apricot-plum parentage.

Plumcot

Variety	Where	Pollination	Notes
Northern Sunset	Row 4, Trees 1, 2, 4 and 7	Partially self-fertile	A plum × apricot hybrid selected for cold hardiness — among the hardiest plumcots available, which is why four were planted. The fruit combines plum juiciness with apricot aroma. Row 4 gives it both parent species to work with: Shiro Gold and Yellow Egg in the same row, and the four apricots one row north. Four together also pollinate one another.

Hazelnut

Variety	Where	Pollination	Notes
Tonda	Row 2, Trees 7 and 10	Wind-pollinated; partner with Gene	A European hazelnut of the Tonda group — the Italian round-nut types grown for the confectionery trade, prized for thin shells and excellent kernel flavour. Hazelnuts are wind-pollinated and dichogamous (male and female flowers on the same plant mature at different times), so two cultivars are needed; Tonda and Gene are planted alternately for exactly that reason. Watch annually for Eastern filbert blight (Section 5.6).
Gene	Row 2, Trees 8 and 9	Wind-pollinated; partner with Tonda	A Canadian selection, recorded in the field notes as “Genie”. Selected for hardiness and blight tolerance in the Ontario climate. Paired alternately with Tonda so pollen moves between them on the wind in late winter, well before the fruit trees stir. A third cultivar can be added over time to widen the pollen window.

Pawpaw

Variety	Where	Pollination	Notes
Pawpaw	Row 5, Trees 8, 9 and 10	Cross-pollinated; hand pollination helps	*Asimina triloba* — native to Ontario, the northern edge of its range reaching the Carolinian zone around Sarnia and the north shore of Lake Erie, and the largest edible fruit native to Canada. Custard-textured flesh tasting of banana, mango and vanilla; it does not ship or store, which is why you never see it in shops and exactly why it is worth growing. The three here pollinate one another as the parent tree does. Bloom peaks around the third week of May, and the maroon flowers are pollinated by flies and beetles rather than bees — moving pollen between plants with a soft brush markedly improves the crop. Harvest late September into October, when the fruit softens and drops; it will not ripen off the tree.

Butternut

Variety	Where	Pollination	Notes
Butternut	South of the parking area	Wind-pollinated	*Juglans cinerea*, a native Ontario tree and **endangered** here, most of the wild population having been lost to butternut canker — so a healthy planted specimen is a genuine contribution. Sweet, oily, richly flavoured nuts. Like all walnut-family trees it produces juglone in its root zone, which suppresses some other plants, so its generous separation from the tree rows is deliberate. Check it annually for canker.

18.2 Berry hedge

Fifty-seven plants of eleven varieties along the south and east fences. All are self-fertile except where noted, so none depends on the orchard for pollination — though the orchard benefits from them, since between them they flower from early spring into autumn and hold a pollinator population on the site.

Kind	Variety	No.	Notes
Gooseberry	Pixwell	10	A prairie-bred variety (North Dakota, 1932) chosen for two practical virtues: it is **nearly thornless** compared with European gooseberries, and the berries hang on long stems well away from the branch, so picking is quick rather than bloody. Very hardy, self-fertile, and resistant to mildew. Pick green for pies and preserves, or leave to ripen pink and sweet for fresh eating. Renewal-prune each winter, removing the oldest third of the stems.
Raspberry	Heritage (everbearing)	10	The standard everbearing red raspberry, from Cornell in 1969. It crops twice — a light summer crop on last year's canes and a heavier main crop in late summer and autumn on the current season's growth. That autumn crop is the point: it extends fresh picking well past the summer raspberries. Firm, well-flavoured, freezes well. Self-fertile. It can be managed the simple way — mow the whole planting to the ground in late winter and take the autumn crop only, which removes all cane-disease carryover in one pass.
Raspberry	Latham	10	An old Minnesota summer-bearing variety from 1914 and still the benchmark for cold-climate hardiness — it survives winters that kill other raspberries. Crops once, in midsummer, on second-year canes. Large firm red berries, excellent for jam and freezing. Self-fertile. Together with the Heritage it gives two separate raspberry harvests, midsummer and autumn.
Strawberry	—	10	Self-fertile and cropping in the first or second year. Renovate after fruiting — shear off the old foliage and thin the runners to about 15 cm apart, which resets the bed for the following season. Replant a fresh bed every three to four years as vigour declines; the runners from the existing plants give you the new plants free.
Blackberry	Black Satin	4	**Thornless**, vigorous and very productive, with large glossy berries ripening early to mid-season. Semi-erect, so it wants the fence for support. Self-fertile. Fruit is on the tart side until fully black-ripe — wait for the shine to dull.
Blackberry	Chester	3	**Thornless** and the hardiest of the three blackberries here, which is why it earns its place — Chester tolerates cold better than most semi-erect types. Ripens late, firm, sweet, and holds well after picking. Self-fertile.
Blackberry	Triple Crown	3	**Thornless**, and widely considered the best-flavoured of the semi-erect blackberries — large, sweet, aromatic berries on a very productive plant. Ripens mid to late season, between Black Satin and Chester. Self-fertile. The three varieties together spread the blackberry harvest across about six weeks.
Currant	American black currant	2	*Ribes americanum* is the **native North American black currant**, not a red currant — worth knowing when reading the label. Hardy, self-fertile, and the fruit is high in vitamin C, intensely flavoured, and best cooked into jam, cordial or syrup rather than eaten raw. Also a good early nectar source for bees. Renewal-prune each winter.
Elderberry	American	2	*Sambucus canadensis*, native to Ontario. Big flat creamy flower heads in midsummer followed by heavy trusses of small dark berries. **Cook the berries — they should not be eaten raw**, and the flowers make cordial. Partially self-fertile but sets far more heavily with a second plant nearby, which is why two were planted. Remove canes older than three years each winter.
Goji berry	—	2	*Lycium barbarum*, self-fertile, cropping on the current season's growth from midsummer well into autumn — the last thing still fruiting in the planting. Prune hard in late winter to a framework about 1 m high, and remove suckers, or it will wander. Small orange-red berries, best dried or cooked.
Saskatoon	Serviceberry	1	*Amelanchier alnifolia*, native, extremely hardy, and **self-fertile** — which matters, because only one was planted. Blueberry-like fruit with a distinctive almond note from the seeds, excellent in pies and preserves. It earns its place twice over: Saskatoon is a **pome fruit** in the same family as apple and pear, so include it in the June fire-blight walk and the dormant spray — and its bloom is the traditional signal to start watching for tent caterpillars (Section 8.1), an indicator now growing where it can be seen from the gate.

19. Glossary of Orchard Terms

Plain-language definitions of the terms used in this guide. Each is also explained where it first appears in the text.

Term	Meaning
Alternate host	A second, unrelated plant that a fungus needs in order to complete its life cycle. Removing one host breaks the cycle.
Btk	*Bacillus thuringiensis kurstaki* — a naturally occurring soil bacterium that affects caterpillars only. Harmless to bees, birds, people and every other insect.
Bud break	The point in spring when buds open and the first leaves emerge. Many sprays are timed relative to it.
Caliper	The diameter of the trunk, measured 15 cm above the ground. Used to size watering and fertilizer rates.
Canker	A dead, sunken patch of bark where a disease has entered the wood.
Central leader	A training system with one dominant upright stem and tiers of side branches around it. Used for apple and pear.
Cross-pollination	Pollen moving from one variety to a different variety. Required by most sweet cherries and pears.
Dormant	The leafless winter period when a tree is not growing. The safest time for heavy pruning and for oil and lime-sulphur sprays.
Drip line	The circle on the ground directly beneath the outermost tips of the branches. Most feeding roots sit near it, so this is where fertilizer goes.
Dwarf / semi-dwarf	Rootstocks that hold a tree to a fraction of its natural size — handy for picking and spraying, but usually needing permanent support.
Floricane	A berry cane in its second year. It fruits, then dies and is cut out.
Girdling	Bark chewed or damaged right around a trunk, severing the connection between roots and canopy. Fatal to the tree.
Graft union	The swelling low on the trunk where the fruiting variety was joined to the rootstock. Anything sprouting below it is rootstock and should be removed.
Green tip	The moment the first green leaf tissue shows at an apple bud. The last safe point for a dormant spray.
Grafting	Joining a shoot of one variety onto an established tree so that it grows on as part of it — the quickest way to add a variety, since the existing roots and framework carry it.
Grafting tape	A stretchable tape binding a graft, holding the cut surfaces together and stopping the scion drying out. Slit it before it constricts the growing wood.
Headland	The clear strip at the ends of the rows, kept free of trees for turning equipment.
Inoculum	The reservoir of fungal spores — in fallen leaves, mummified fruit or cankers — that starts the following season’s infection.
IPM	Integrated pest management: watch the orchard first, then use the least disruptive treatment that will do the job, rather than spraying to a fixed calendar.
Juglone	A natural compound released around the roots of walnut-family trees that suppresses the growth of some other plants.
Mummified fruit	Shrivelled, dried fruit left hanging in the tree or lying beneath it. The main way brown rot survives the winter.
OMRI-listed	Reviewed and accepted for use in certified organic growing.
Open centre (vase)	A training system with no central stem — three to five limbs radiating outward, leaving the middle open to light and air. Used for stone fruit.
Petal fall	When the blossom petals drop. The single most important spray timing for apple and pear.
Hardiness zone	A rating of average winter minimums. This site is roughly zone 5b — a full zone colder than Niagara, which is where most Ontario fruit advice is written for.
Phenology	Using one plant’s growth stage as a calendar for another — for example, serviceberry bloom signalling tent caterpillar season.
Pink bud	Flower buds showing colour but not yet open. A copper-spray timing for fire blight.
Pollinizer / pollinator	The *pollinizer* is the tree supplying the pollen. The *pollinator* is the bee or other insect that carries it.
Pome fruit	A fleshy fruit built around a core — apple, pear, crabapple, Saskatoon.

Term	Meaning
Primocane	A first-year berry cane. Next season it becomes a floricane and fruits.
Renewal pruning	Removing the oldest stems of a bush each year so it continually replaces itself with young, productive wood.
Renovation	For strawberries: shearing the foliage off and thinning the runners after harvest, which resets the bed for the following year.
Rootstock / scion	The *rootstock* is the root system a variety is grafted onto and controls the tree's eventual size. The *scion* is the fruiting variety on top.
S-allele	A genetic tag carried in cherry pollen. If it matches the tag in the flower it lands on, the flower rejects it, so varieties in the same group cannot pollinate each other.
Scaffold	One of the main structural limbs of a tree, established in the first few years.
Self-fertile	A variety that sets fruit with its own pollen. One tree on its own will crop.
Self-unfruitful	A variety that needs pollen from a different variety in order to set fruit.
Spur	A short, stubby shoot that carries flower buds year after year. Pruning to spurs keeps fruiting concentrated and within reach.
Stone fruit	A fruit with a single hard pit — cherry, plum, apricot, peach, nectarine, plumcot.
Stromata	Small raised black bumps arranged in rows along a stem — the fruiting bodies of Eastern filbert blight on hazelnut.
Sucker	A shoot growing from the rootstock below the graft union. Always remove it.
Topworking	Grafting new varieties onto the branch framework of a mature tree rather than onto a young rootstock.
Waterlogged	Soil so wet that the air is driven out of it. Roots cannot breathe, and the tree wilts and yellows in a way easily mistaken for drought.
Sunscald	Bark splitting on the south-west side of a trunk, caused by late-winter sun warming it by day and it refreezing at night. Prevented by painting the trunk white.
Thinning (fruit)	Removing surplus fruitlets after petal fall so the rest size up properly and do not touch each other, which reduces rot.
Triploid	A variety carrying three sets of chromosomes instead of the usual two. Its own pollen is largely unusable, so it receives pollen rather than supplying it.
Vigour	How strongly a tree grows. Governed mostly by rootstock, then by soil and pruning.
Water sprout	A vigorous vertical shoot growing straight up from a branch. Unproductive — remove it.

Georgina Island Orchard Plan · Chris Gynan RPF, Certified Arborist · July 2026. Layout scaled from aerial photography at 12.7 px/m; fenced area 3,718 m^2 (0.37 ha). Species and varieties as recorded in the field.